

Deep Learning-Based Image Reconstruction for CT Scans in Radiation Therapy

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Introduction

Radiation Therapy

A vital treatment approach for cancer patients, utilizing targeted radiation to destroy cancer cells while minimizing harm to healthy tissues.

Dose Distribution

Dose distribution refers to the spatial distribution of radiation doses within the target area and surrounding tissues during radiation therapy.

Current challenges

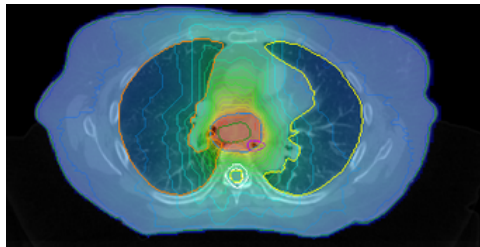
Dose prediction in radiation therapy relies on time-consuming Monte Carlo simulation, which can take several days to yield accurate results.

Objective

- Develop a deep learning model to efficiently compute and predict the dose distribution in radiation therapy.
- Utilize CT scan images of prostate cancer to accurately predict the dose distribution.

Computed Tomography (CT) Scan

A CT scan is a medical imaging technique that uses X-rays and computer processing to create detailed cross-sectional images of the body [Bushberg et al, 2011].



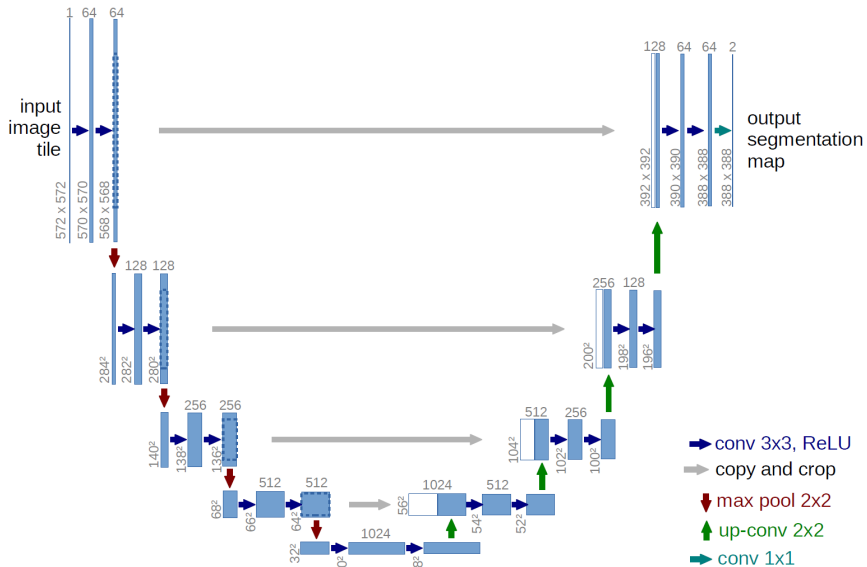
Deep Learning

Deep learning is a subset of machine learning that focuses on training deep neural networks with multiple layers to learn complex patterns and representations from data.



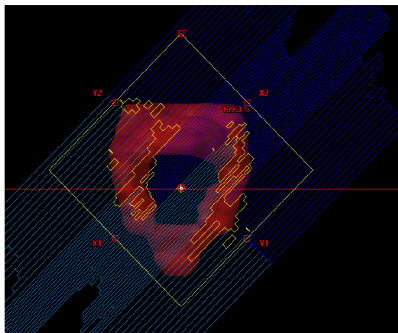
U-Net Architecture

- U-Net is a widely used convolutional neural network (CNN) architecture for image reconstruction tasks [Ronneberger et al, 2015].
- It is specifically designed for tasks that involve both encoding and decoding of image features, making it suitable for image segmentation and reconstruction.

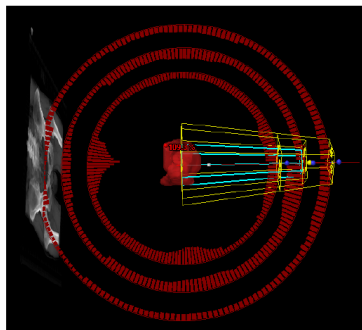


- Consists of CT scans in DICOM (.dcm) format and corresponding fluence data in MetalImage (.mhd) format for eight prostate cancer patients.
- CT scans provide detailed anatomy images, while fluence data captures radiation beam intensity distribution.

- Dataset obtained using VMAT (Volumetric Modulated Arc Therapy) technique on the Clinac 2100 C machine.
- VMAT delivers radiation from multiple angles, enabling precise and targeted radiation delivery.
- Each patient has 178 fluence maps per arc, revealing radiation dosage in different prostate regions.

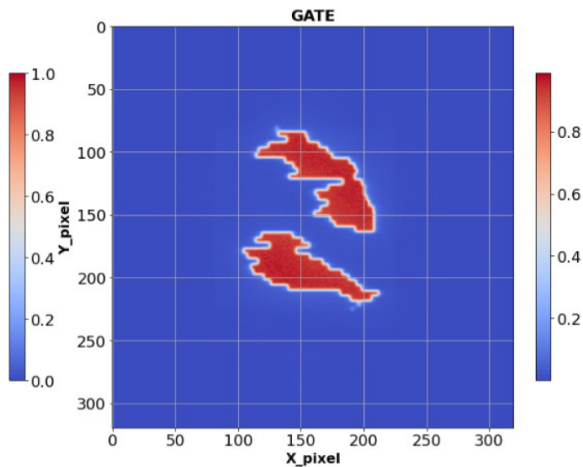
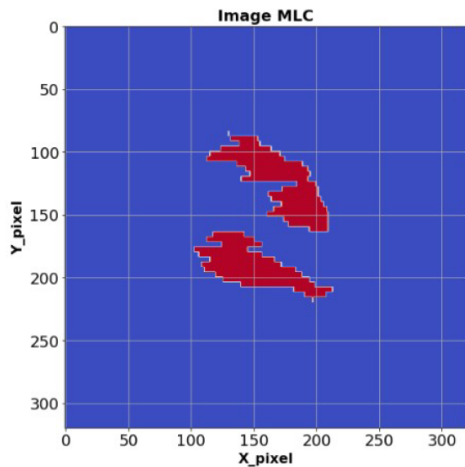


Beam eye view



Visualisation of 3 arcs

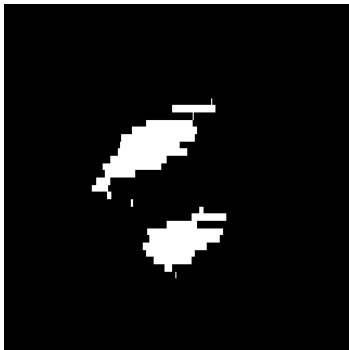
An already written python script was used to process the .dcm & .mhd files into the following images



- Our U-Net model, implemented using TensorFlow, follows the U-Net architecture, utilizing contracting and expanding paths.
- The final layer of the U-Net model uses a 1×1 convolution followed by a sigmoid activation function to generate the segmentation mask.

Data preparation

- The data preparation step involves loading images, resizing them to a consistent size of 320x320 pixels, converting to grayscale, normalizing pixel values, and splitting the dataset into training and validation (80/20)



CT scan



Dose distribution

- The model is trained on the training set and evaluated on the validation set.
- The performance is assessed using validation loss and accuracy.
- reconstructed images are saved for visualisation

Validation Loss measures the discrepancy between the predicted outputs and the actual ground truth values during the validation phase.

Validation Accuracy is the proportion of correctly predicted instances out of the total number of instances in the validation set.

Model Output

- The trained model is saved under model.h5 & logs
- Reconstructed images (= dose distribution)

Results

- There was no significant results for the most part.
- The model was running correctly but the reconstructed images were just black images with no relevant information.



Reconstructed dose distribution

However, just yesterday we got some promising results.



CT scan



Reconstructed Dose distribution

- This shows that our network is at somewhat working.
- further testing and fine-tuning of are still required to monitor the accuracy.
- In Theory, The reconstructed dose distribution can be utilized to extract fluence maps, which capture the spatial distribution of radiation beam intensities during treatment.

- Fine tuning the model parameters to improve it's accuracy.
- extracting the relevant fluence data to obtain the beam parameters (angle, intensity etc..).
- Test the model on a new dataset of different cancer type/forms.

- Developed a deep learning model based on U-Net architecture for dose distribution prediction in radiation therapy.
- Promising results obtained, particularly in recent experiments.
- Future work includes fine-tuning the model parameters, extracting relevant fluence data for beam parameters, and testing on diverse cancer datasets.

References



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The End

Questions? Comments?